Sahaj Bhadja

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Summary

Software engineer with 2+ years building FastAPI services, React/Next.js interfaces, REST APIs, and AI-enhancement pipelines. Experienced in Python, TypeScript, PostgreSQL, RAG, prompt engineering, structured outputs, Docker, Kubernetes, pytest, CI/CD, and cloud deployment.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built FastAPI endpoints, Azure SQL services, and React dashboards for 500+ users, optimizing indexed queries and backend access paths to reduce production latency from 1.2 seconds to 350 milliseconds.
- Orchestrated 5,000+ monthly jobs through explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving API observability, operational resilience, recovery, maintainability, and systematic production issue diagnosis.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask REST services handling 10,000+ daily requests, integrating OpenAI, Anthropic, and Qwen providers for document grading while resolving production defects affecting 150+ users.
- Created multithreaded processing for 1,000+ files with structured validation, AWS RDS persistence, progress monitoring, and failure recovery, reducing turnaround 60% while maintaining reliable behavior under concurrent production load.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built testable Python and PyTorch pipelines processing 100,000+ trajectory logs, improving planning accuracy 18% through reproducible experiments, structured model evaluation, performance monitoring, validation checks, and systematic debugging.
- Developed a spatial-temporal encoder combining map geometry and agent context through multi-head attention, improving closed-loop stability 15% while documenting experiments and validating outputs across diverse simulation scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services and scalable REST integrations using filtering, pagination, caching, AWS Glue, S3, and SQL, improving production response times 30% and data-processing efficiency 68% across cloud applications.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito, accelerating technical troubleshooting, release verification, and reliable cloud application deployments across Windows, Linux, and macOS.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js services across eight microservices with authentication, role-based access, WebSocket messaging, and reusable interfaces, processing 2 million+ Azure SQL records through documented APIs.
- Automated Docker builds, functional tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving service reliability, deployment consistency, and operational visibility.

Skills

Frontend & Languages: Python, JavaScript, TypeScript, Java, SQL, React, Next.js, HTML5, CSS3, functional components, responsive interfaces, reusable components
Backend & AI: FastAPI, Flask, Node.js, REST APIs, microservices, OpenAI, Anthropic, Qwen, LLaMA-2, RAG, FAISS, embeddings, semantic search, prompt engineering, structured outputs
Data & Cloud: PostgreSQL, Azure SQL, MySQL, MongoDB, AWS RDS, AWS, Azure, S3, AWS Glue, Docker, Kubernetes, schema design, query optimization, caching
Quality & Delivery: Git, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, functional testing, automated testing, validation, debugging, logging, monitoring, code review, documentation, Agile, Scrum

Projects

Big-Brain | Live | *LLaMA-2, RAG, FAISS, PostgreSQL, REST APIs, Clerk*
- Built an authenticated RAG application processing 50,000+ unstructured documents with LLaMA-2, FAISS, PostgreSQL, and REST APIs, combining semantic retrieval, structured validation, access controls, and real-time collaboration workflows.

Chat2SQL | Code | *Java, TypeScript, SQL, PostgreSQL, MySQL, MongoDB, Docker*
- Built an AI-powered SQL workspace supporting 10+ databases with schema-aware query generation, structured validation, optimized execution, metadata exploration, and Docker-backed deployment across modular Java and TypeScript services.

PiXMod | Live | *Next.js, TypeScript, MongoDB, Stripe, CLIP, SAM*
- Built a full-stack AI application using Next.js, TypeScript, CLIP, and SAM, implementing reusable interfaces, structured image processing, secure authentication, MongoDB persistence, Stripe integration, and responsive cross-device workflows.